

9. Ascorbic acid,  $C_6H_8O_6$ , is the main component of vitamin C tablets. Its name is derived from *a-* (meaning “no”) and *scorbutus* (scurvy), the disease caused by a deficiency of vitamin C. A student was asked to find the percentage of ascorbic acid in identical vitamin C tablets.

She was told to use the following method.

- Fill a burette with  $0.100 \text{ mol dm}^{-3}$  sodium hydroxide solution.
- Weigh a conical flask and record its mass.
- Add a vitamin C tablet to the flask, reweigh it and record its mass.
- Add about  $50 \text{ cm}^3$  of deionised water to the flask and swirl to break up the tablet.
- Heat the flask gently for 5 to 10 minutes.
- After the solution has cooled add a few drops of a suitable indicator.
- Carry out a rough titration of this solution with the sodium hydroxide solution.
- Accurately repeat the procedure several times and calculate a mean titre.

- (a) A **three** decimal place balance was used. The mass of each vitamin C tablet was 500 mg.

Calculate the maximum percentage error in the weighing of the tablet.  
You **must** show your working.

[2]

Maximum percentage error = ..... %

- (b) (i) Suggest why she did not need to measure the volume of water accurately. [1]

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- (ii) Suggest why she heated the flask for 5 to 10 minutes. [1]

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- (c) The student used the results from three titrations to calculate a mean titre. Some of her results are shown below.

| Titration                         | 1     | 2     | 3    |
|-----------------------------------|-------|-------|------|
| Final reading / cm <sup>3</sup>   | 26.90 | 26.90 |      |
| Initial reading / cm <sup>3</sup> | 0.25  | 0.15  | 0.20 |
| Titre / cm <sup>3</sup>           | 26.65 | 26.75 |      |

Mean titre = 26.73 cm<sup>3</sup>

Determine the **final reading** for the third titration.

[2]

Final reading = ..... cm<sup>3</sup>

- (d) Ascorbic acid can decompose upon exposure to air. If this reaction occurred before the titration was completed, state how it might affect the titration results. Explain your answer. [2]

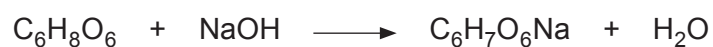
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- (e) The equation for the reaction between ascorbic acid and sodium hydroxide is given below.



$M_r$  176

The percentage of ascorbic acid is identical in each 500 mg tablet. Calculate the percentage of ascorbic acid in each vitamin C tablet. [3]

Percentage ascorbic acid = ..... %

- (f) Sulfuric acid and hydrochloric acid are strong acids.

- (i) Calculate the pH of a solution of  $0.010 \text{ mol dm}^{-3}$  sulfuric acid,  $\text{H}_2\text{SO}_4$ . [2]

pH = .....

- (ii) When hydrochloric acid is heated with  $\text{MnO}_2$  it reacts according to the following equation.



Explain why this can be classified as a redox reaction. [2]

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